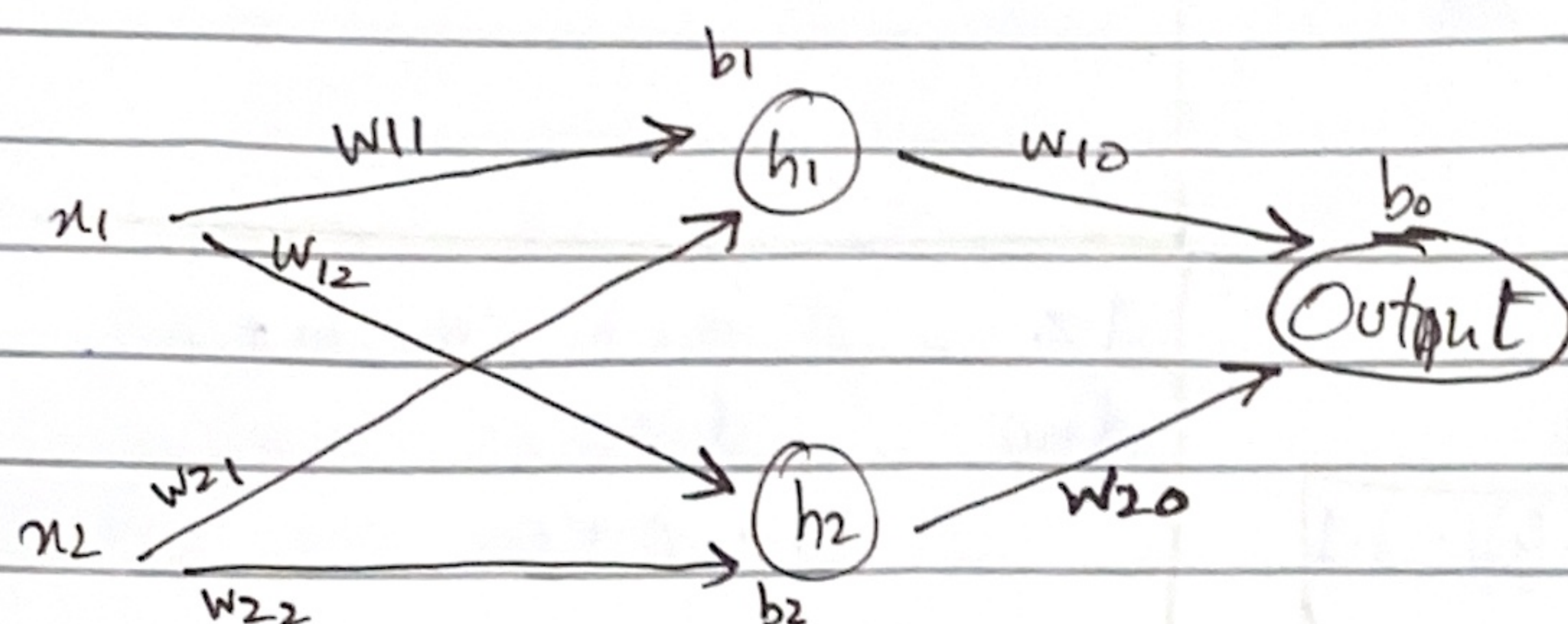


LECTURE-19

Wednesday
15/4/26

Q: $x_1=1, x_2=1, y=1$, $\alpha=0.2$, Activation = Sigmoid
All weights = 0.1 except $w_{20}=1$, All Bias = 0.



$$h1 = \sigma(x_1 \cdot w_{11} + x_2 \cdot w_{21} + b_1) = \sigma(0.1 + 0.1 + 0) = 0.55$$

$$h2 = \sigma(x_1 \cdot w_{12} + x_2 \cdot w_{22} + b_2) = \sigma(0.1 + 0.1 + 0) = 0.55$$

$$\text{output} = \sigma(h1 \cdot w_{10} + h2 \cdot w_{20} + b_0) = \sigma[(0.55) \cdot (0.1) + (0.55) \cdot (1) + 0] = \boxed{0.6467}$$

$$\text{Error} = y - \text{output}$$

$$= 1 - 0.6467 = \boxed{0.353}$$

• $\Delta \text{output} = \text{Error} \cdot \sigma'(\text{output})$ Sigmoid derivative
 $\sigma' = \text{output} (1 - \text{output})$

$$\Delta o = 0.353 \cdot 0.6467 (1 - 0.6467) = \boxed{0.08}$$

$$\begin{cases} \text{new: } w_{10} = w_{10} + \alpha (\Delta \text{output}) (h1) = 0.1 + (0.2)(0.08)(0.55) = \boxed{0.1088} \\ \text{new: } w_{20} = w_{20} + \alpha (\Delta \text{output}) (h2) = 0.1 + (0.2)(0.08)(0.55) = \boxed{1.0088} \\ \text{new: } b_0 = b_0 + \alpha (\Delta \text{output}) = 0 + 0.2 + 0.08 = \boxed{0.016} \end{cases}$$

Same as formula derived from delta Rule (in prev. lecture)

★ we've found new w_{10} & w_{20} by moving back from output towards h_1 & h_2 .

★ Now we'll find ~~new~~ ~~new~~ Δh_1 and Δh_2 (like Δoutput) so we can later change weights of hidden layer (i.e. $w_{11}, w_{12}, w_{21}, w_{22}$)

⇒ For Δh_1 we need; derivative of $\sigma'(h_1)$, w_{10} , Δoutput

$$\sigma'(h_1) = h_1(1-h_1)$$

$$\sigma'(h_2) = h_2(1-h_2)$$

$$\bullet \Delta h_1 = \sigma'(h_1) \cdot w_{10} \cdot \Delta \text{output} = \boxed{0.002}$$

$$\bullet \Delta h_2 = \sigma'(h_2) \cdot w_{20} \cdot \Delta \text{output} = \boxed{0.002}$$

⇒ Now we'll use Δh_1 & Δh_2 to find $w_{11}, w_{12}, w_{21}, w_{22}$.

new:

$$w_{11} = w_{11} + \alpha \cdot \Delta h_1 \cdot x_1 = 0.1 + (0.2 * 0.002 * 1) = \boxed{0.1004}$$

$$w_{12} = w_{12} + \alpha \cdot \Delta h_2 \cdot x_1 = 0.1 + (0.2 * 0.02 * 1) = \boxed{0.104}$$

$$w_{21} = w_{21} + \alpha \cdot \Delta h_1 \cdot x_2 = \boxed{0.1004}$$

$$w_{22} = w_{22} + \alpha \cdot \Delta h_2 \cdot x_2 = \boxed{0.104}$$

$$b_1 = b_1 + \alpha \cdot \Delta h_1 = 0 + 0.2 * 0.002 = \boxed{0.0004}$$

$$b_2 = b_2 + \alpha \cdot \Delta h_2 = 0 + 0.2 * 0.02 = \boxed{0.004}$$

⇒ After new weights and bias till we reached inputs (x) using Back Prop. Now we'll use new biases (b_0, b_1, b_2) and new weights ($w_{11}, w_{12}, w_{21}, w_{22}, w_{10}, w_{20}$) to do "forward Pass" i.e.: find h_1, h_2 and then new 'y' (Output) like we did on first step. This will complete a cycle where we did Forward Pass and Back Propagation.

Then we'll again compute error ($y - \Delta y$) & do back prop. to repeat.

Practically in code: we'll keep repeating till a constant or almost constant change is observed after some iterations that we'll keep increasing.

Q: IF we'll have two neurons in output layer having "O1" and "O2" instead of single output:

weight b/w h1 and O1

weight b/w h1 and O2.

$$\Delta h1 = \sigma'(h1) \cdot (\Delta O1) \cdot W_{101} + \sigma'(h1) \cdot \Delta O2 \cdot W_{102}$$

$\Delta h2 =$ same type.

$$\Delta O1 =$$

$\Rightarrow$ No. of neurons in output layer = No. of Bits in output (y) in One-hot-Encoding form.

Therefore since we're O1 & O2 (two neurons in output layer);

so we'll have $y = \begin{bmatrix} 0 & 1 \end{bmatrix}$ or $\begin{bmatrix} 1 & 0 \end{bmatrix}$

O1 O2

O1 O2